Network Planning and Strategy Development

Network planning refers to the process of designing, organizing, and implementing a computer network infrastructure that meets the communication needs of an organization efficiently and securely. Proper network planning ensures that networks support business operations, allow future expansion, and minimize operational costs.

A well-planned network improves:

- Communication efficiency
- Data sharing
- System reliability
- Security of organizational information
- Scalability for future growth

Without proper planning, organizations may face network congestion, frequent downtime, security breaches, and high maintenance costs.

Network Planning Approaches

Organizations adopt different approaches when planning networks depending on their needs, size, and available resources.

1 Top-Down Network Planning Approach

In this approach, **network design begins with organizational requirements and business objectives** before selecting technologies.

Steps involved:

1. Identify business goals and requirements
2. Determine application needs
3. Design logical network architecture
4. Select hardware and software
5. Implement and test the network

Advantages:

- Aligns the network with business objectives
- Ensures user needs are prioritized

- Reduces unnecessary technology investments

Disadvantages:

- Requires detailed analysis and planning
- May take longer to implement

2 Bottom-Up Network Planning Approach

This approach begins by **examining existing network technologies and infrastructure**, then building the network based on available technology.

Steps involved:

1. Assess existing hardware and software
2. Evaluate available network technologies
3. Expand or modify the network
4. Integrate new components

Advantages:

- Faster to implement
- Uses existing infrastructure

Disadvantages:

- May not fully meet organizational needs
- Can result in inefficient network structures

3 Hybrid Network Planning Approach

This approach combines **top-down and bottom-up approaches**.

Characteristics:

- Business needs are analyzed first
- Existing technologies are also considered
- New technologies are integrated gradually

Advantages:

- Balanced and flexible
- Cost-effective
- Supports long-term network growth

Planning Network Strategies

Network strategy planning focuses on long-term goals and the direction of network development in an organization.

Key components include:

1 Assessing Organizational Requirements

Organizations must determine:

- Number of users
- Types of applications used
- Data traffic requirements
- Security needs
- Remote access requirements

Example:
A university network requires support for:

- E-learning platforms
- Email services
- Student information systems
- Research databases

2 Network Capacity Planning

Capacity planning ensures the network can **handle current and future traffic loads**.

Factors considered include:

- Bandwidth requirements
- Number of connected devices

- Application usage
- Peak network traffic

Proper capacity planning prevents:

- Network congestion
- Slow data transmission
- System downtime

3 Network Security Strategy

Organizations must develop strategies to protect network resources.

Key security components include:

- Firewalls
- Intrusion detection systems
- Access control policies
- Data encryption
- Authentication mechanisms

Benefits:

- Protects sensitive organizational data
- Prevents cyberattacks
- Ensures system integrity

4 Network Scalability Planning

Scalability ensures the network can **grow as the organization expands**.

Examples:

- Adding more users
- Expanding branches
- Introducing new services

Scalable networks allow:

- Easy hardware upgrades
- Additional network segments
- Integration of emerging technologies

Organizational Networking Policies

Network policies are rules and guidelines that govern how network resources are used, accessed, and managed within an organization.

They help maintain security, efficiency, and accountability.

1 Acceptable Use Policy (AUP)

Defines how employees can use organizational network resources.

Typical rules:

- No unauthorized software installation
- No illegal downloads
- No access to restricted websites
- Proper use of email systems

2 Security Policy

Defines measures for protecting network infrastructure.

Examples:

- Password complexity requirements
- User authentication procedures
- Access privileges
- Data backup policies

3 Remote Access Policy

Defines rules for accessing the network remotely.

Includes:

- VPN usage
- Secure login procedures

- Multi-factor authentication

4 Data Protection Policy

Ensures confidentiality and integrity of organizational data.

Includes:

- Data encryption
- Backup procedures
- Access restrictions
- Data retention policies

5 Network Documentation and Standards

Network documentation provides detailed records of network configurations, components, and procedures.

Proper documentation helps:

- Network troubleshooting
- System maintenance
- Network upgrades
- Staff training

Types of Network Documentation

1. Network Topology Diagrams

Visual representation of network structure showing:

- Routers
- Switches
- Servers
- Network connections

2. IP Address Documentation

Records all assigned IP addresses and devices.

3. Configuration Documentation

Contains:

- Router configurations
- Switch settings
- Firewall rules

4. Maintenance Procedures

Includes procedures for:

- Network upgrades
- Backup systems
- Fault troubleshooting

Network Standards

Network standards ensure compatibility and interoperability between devices and technologies.

Examples include standards developed by:

- Institute of Electrical and Electronics Engineers (IEEE)
- International Organization for Standardization (ISO)
- Internet Engineering Task Force (IETF)

Examples of standards:

- Ethernet (IEEE 802.3)
- Wireless networks (IEEE 802.11)
- Internet protocols

Benefits of standards:

- Interoperability
- Reliable communication
- Vendor compatibility